

CS23431 DESIGN AND ANALYSIS OF ALGORITHMS

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UNIT I INTRODUCTION

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Notion of an Algorithm – Fundamentals of Algorithmic Problem Solving – Important Problem Types – Fundamentals of the Analysis of Algorithmic Efficiency – Asymptotic Notations and their properties. Analysis Framework – Empirical analysis - Mathematical analysis for Recursive and Non-recursive algorithms – Visualization.

UNIT II BRUTE FORCE AND DIVIDE-AND-CONQUER

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Brute Force – Computing an – String Matching – Closest - Pair and Convex - Hull Problems - Closest - Pair and Convex - Hull Problems - Knapsack Problem – Assignment problem. Divide And Conquer Methodology – Binary Search – Merge sort – Quick sort – Heap Sort – Multiplication of Large Integers – Closest - Pair and Convex - Hull Problems.

UNIT III DYNAMIC PROGRAMMING AND GREEDY TECHNIQUE

9

Dynamic programming – Principle of optimality - Coin changing problem, Computing a Binomial Coefficient – Floyd's algorithm – Multi stage graph - Optimal Binary Search Trees - Knapsack Problem and Memory functions. Greedy Technique – Container loading problem - Prim's algorithm and Kruskal's Algorithm – 0/1 Knapsack problem, Optimal Merge pattern - Huffman Trees.

UNIT IV ITERATIVE IMPROVEMENT

9

The Simplex Method - The Maximum-Flow Problem – Maximum Matching in Bipartite Graphs, Stable marriage Problem.

UNIT V COPING WITH THE LIMITATIONS OF ALGORITHM POWER

9

Lower - Bound Arguments - P, NP NP- Complete and NP Hard Problems. Backtracking – n-Queen problem - Hamiltonian Circuit Problem – Subset Sum Problem. Branch and Bound – LIFO Search and FIFO search - Assignment problem – Knapsack Problem – Travelling Salesman Problem - Approximation Algorithms for NP-Hard Problems – Travelling Salesman problem – Knapsack problem.

TOTAL : 45 PERIODS

TEXTBOOKS:

1. Anany Levitin, – Introduction to the Design and Analysis of Algorithms, Third Edition, Pearson Education, 2012. (UNIT I)
2. Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, Computer Algorithms/C++, Second Edition, Universities Press, 2007. (UNIT II, III, IV & V).

REFERENCES:

- 1 Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, – Introduction to Algorithms, Third Edition, PHIL earning Private Limited, 2012.
- 2 Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, – Data Structures and Algorithms, Pearson Education, Reprint 2006.
- 3 Harsh Bhasin, – Algorithms Design and Analysis, Oxford university press, 2015.
- 4 <http://nptel.ac.in/>